

Krishi Rajeshkumar Shah

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PROFILE

Final year Honours Computer Science student at York University building applied machine learning and generative AI systems end to end, from data validation through training, calibration, explainability and deployment. One year of production software engineering with the Ontario Government (Co-op of the Year nominee), a four month formal research role on AI generated image detection, and a portfolio of open source ML projects spanning computer vision, retrieval augmented generation, clinical risk modeling and fairness auditing. Published author of *The Awakening Code* and technical writer across eight publications. Comfortable working in agentic development environments including Claude Code, OpenAI Codex, Cursor and GitHub Copilot.

EDUCATION

York University, Lassonde School of Engineering

Bachelor of Science, Computer Science (Honours)

Toronto, ON

Expected Aug. 2027

- **Machine learning and data:** Applied Machine Learning, Statistics for Data Science, Statistical Modeling, Data Mining, Big Data Analytics, Database Systems (EECS 3421), Information Networks (EECS 4414).
- **Core systems and theory:** Operating Systems (EECS 3221), Professional Practice in Computing (EECS 3000), Advanced Object Oriented Programming, Algorithms and Data Structures.
- **Research:** EECS 4080 Research Project in Computer Science, supervised by Prof. Mona Nasery.

Parul University

Diploma in Computer Engineering, GPA 3.9/4.0

Gujarat, India

2020 – 2023

- 5x University Gold Medalist for highest GPA overall; Dean's List.

TECHNICAL SKILLS

Languages and query: Python, SQL, R, Java, C++, Bash, PySpark

Machine learning and deep learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Random Forest, Gradient Boosting, SVM, CNNs, EfficientNet, transfer learning, timm, Hugging Face Transformers

LLMs, RAG and agents: LangChain, FAISS, BM25 hybrid retrieval, cross encoder reranking, vector search, LLM tool calling, prompt engineering, GPT-3.5-Turbo, BERT, FinBERT, all-MiniLM-L6-v2, agentic workflows

Statistics and analysis: Exploratory data analysis, hypothesis testing, statistical modeling, regression, classification, feature engineering, cross validation, resampling, error analysis

Evaluation and explainability: ROC-AUC, F1, precision and recall trade off analysis, threshold sweeps, confidence calibration, temperature scaling, expected calibration error, SHAP, Grad-CAM, FFT spectral analysis, t-SNE, Fairlearn, disparate impact, demographic parity, equalized odds

ML systems and engineering: MLflow, FastAPI, Gradio, Docker, CI/CD, GitHub Actions, Azure DevOps, Git, unit testing, pytest, reproducible seeding, model registry, REST APIs

Data engineering and databases: PostgreSQL, schema design, indexing and query optimization, ETL pipelines, data validation, data quality governance, Hadoop, Apache Spark, pandas, NumPy

Visualization and BI: Tableau, Power BI, Matplotlib, Seaborn, executive dashboarding, KPI reporting

Cloud and platforms: Microsoft Azure, Azure DevOps, GitHub, Linux

AI assisted development: Claude Code, OpenAI Codex, Cursor, GitHub Copilot, AI agent frameworks and tool calling workflows

Professional: Technical writing, code review, mentorship, presenting to non technical stakeholders, project scoping

PROFESSIONAL EXPERIENCE

AI Research Assistant

York University, supervised by Prof. Mona Nasery

May 2026 – Aug. 2026

Toronto, ON

- Fine tuned EfficientNet-B3 on 120,000 images to 96.96% test accuracy and 0.9971 AUC-ROC, selecting the configuration from a six run sweep across learning rate and weight decay on a single T4 GPU with AdamW and cosine annealing.
- Measured whether the confidence scores meant anything rather than stopping at accuracy: temperature scaling learned at $T = 1.2189$ brought expected calibration error to 0.0026.

- Designed a resolution matched control experiment that overturned the headline result. 93% of ordinary high resolution photographs came back FAKE, showing the earlier 94% cross generator scores were reading resampling artifacts rather than generator fingerprints. Matched resolution detection fell to between 35% and 62%.
- Evaluated cross generator transfer across five families: StyleGAN, Stable Diffusion 3, Flux, Midjourney v6, Janus-Pro and GPT-4o. Traced the hardest case, GPT-4o at 86.3% detection, through FFT spectra, t-SNE embeddings and Grad-CAM, which ruled out an attention problem and pointed to missing spectral features instead.
- Swept the decision threshold across seven cut points to map the precision and recall trade off for deployment scenarios with different false positive tolerances.
- Wrote 71 unit tests across 8 modules with seeds fixed throughout, so every figure in the final report reruns from scratch. Shipped a Gradio demo with calibrated confidence and Grad-CAM overlays, then disabled public uploads because the control experiment proved real photos would be flagged.
- Delivered across a seven milestone project contract with a final capstone presentation in August 2026.

Software Developer Co-op | Co-op of the Year Nominee

Jan. 2025 – Dec. 2025

Ontario Government, Gouvernement de l'Ontario

Toronto, ON

- Recurring inconsistencies in ministry reporting traced back to unvalidated schema changes reaching the data at ingestion. Rewrote the validation layer in Python and SQL across 10+ workflows and brought reported data errors down by roughly 30%.
- Refactored SQL schemas, queries and indexes behind 5+ enterprise applications, improving query performance 25% and cutting report generation time for teams reading those numbers daily.
- Built automated test suites and release automation in Azure DevOps, reducing production deployment time 40% and moving the team from weekly releases to same day releases.
- Owned parts of the data quality layer that analytics and reporting depend on, the same reliability problem that sits underneath any production machine learning pipeline.
- Stack: Python, SQL, PostgreSQL, Azure DevOps, CI/CD, automated testing, Git.

Software Developer, Analytics

Feb. 2025 – Apr. 2025

All Good Accounting

Toronto, ON

- Built a financial reporting application that replaced a manual Excel based process, delivered through the Level UP industry program.
- Scoped requirements directly with the client and shipped reporting views that non technical staff could operate without developer support.

Data Science Intern

Nov. 2022 – Jan. 2023

NFOX Technologies India Pvt. Ltd.

Gujarat, India

- Compared Random Forest, SVM and Gradient Boosting on 10K+ customer churn records, moving accuracy from 73% to 85%. Feature engineering, not the model choice, produced nearly all of the gain.
- Benchmarked a BERT based text classifier against bag of words baselines using precision, recall and ROC-AUC, tracking each iteration until the 15% lift justified the extra training cost.
- Built Tableau dashboards over KPI and financial performance metrics so business stakeholders could read results without waiting on an analyst.

Data Analytics Intern

May 2022 – Jul. 2022

Suven Consultants and Technology Pvt. Ltd.

Gujarat, India

- Built reusable Python and SQL pipelines for preprocessing and analysis across 5+ datasets, generating insights that informed 3 product decisions.
- Designed 3 interactive Power BI dashboards for KPI tracking and executive reporting, reducing manual reporting effort by 50%.

Data Analyst Intern

Aug. 2022 – Sep. 2022

Code Clause Pvt. Ltd.

Pune, India

- Delivered data analysis and visualization tasks in Python, covering cleaning, exploratory analysis and reporting on assigned datasets.

Software Developer

Apr. 2022 – May 2022

Jeenweb Technologists Pvt. Ltd.

Gujarat, India

- Contributed to web application development and maintenance during an early stage development placement.

SELECTED PROJECTS

- TruePixel: AI Generated Image Detection** [GitHub] | *PyTorch, EfficientNet, Grad-CAM, Gradio* 2026
- EfficientNet-B3 detector fine tuned on CIFAKE with a 1536 to 2 classifier head via timm, 80/20 split at seed 42, best checkpoint at epoch 2. Reached 96.96% accuracy and 0.9971 AUC on a 20,000 image test set.
 - Full evaluation stack: temperature scaling, expected calibration error, threshold sweeps, FFT frequency analysis, t-SNE embedding maps and Grad-CAM attention overlays, plus a resolution matched control that exposed a confound in the standard cross generator protocol.
 - 71 unit tests across 8 modules, fixed seeds, and a Gradio demo integrating the full inference pipeline.
- FinRAG AI** [GitHub] | *LangChain, FAISS, FinBERT, Retrieval Augmented Generation* 2025
- Question answering pipeline over SEC filings and earnings transcripts. The first version looked fine until the retrievals were inspected: naive chunking split answers across boundaries and returned confident, incomplete facts.
 - Hybrid BM25 and FAISS retrieval with an ms-marco cross encoder reranker, all-MiniLM-L6-v2 embeddings, FinBERT for financial sentiment and GPT-3.5-Turbo for generation. Retuning chunk overlap and relevance scoring lifted context recall to 92%.
 - Chose a local FAISS index over managed infrastructure for control over the retrieval layer. GitHub Actions runs the test suite on every commit.
- MedRiskML** [GitHub] | *XGBoost, LightGBM, SHAP, MLflow, FastAPI, Docker* 2025
- Predicts 30 day hospital readmission across 100K+ patient records using 30+ engineered clinical features, reaching 0.87 ROC-AUC after cross validated tuning of XGBoost and LightGBM.
 - Clinical advisors needed to know why one patient was flagged and another was not, so SHAP explanations run per prediction inside the FastAPI service rather than as an offline global summary. MLflow tracked every experiment and registered the winning model; Docker packages the full stack.
- AI Bias and Fairness Explorer** [GitHub] | *Fairlearn, Tableau, Statistical Analysis* 2025
- Measured disparate impact, demographic parity difference and equalized odds across demographic splits with Fairlearn, then surfaced the results in Tableau so non technical reviewers could explore the gaps themselves.
- AI Terminal Assistant** [GitHub] | *LLM Tool Calling, Open Source* 2025
- Converts plain English into shell commands through LLM tool calling, covering pipe chains and conditional logic. MIT licensed and open to contributions.
- Toronto AI and ML Internship Radar** | *Python, GitHub Actions, Web Scraping, Open Source* 2026
- Live feed of Ontario AI and ML internships and co-ops, rescraped hourly through GitHub Actions with no paid APIs or accounts.
 - Classification rules filter by region, recency and hiring cycle, tuned for recall so ambiguous postings are retained rather than silently dropped.
- groundcheck** | *LLM Uncertainty and Hallucination Detection, Open Source* 2026
- Open source library for detecting hallucination and quantifying uncertainty in large language model outputs, packaged for reuse by other developers.
- TruePixel Preprocess C++** | *C++, Image Preprocessing* 2026
- Reimplemented the TruePixel image preprocessing path in C++ to move the inference front end off Python and exercise systems level performance work.
- Healthcare Dashboard** | *Power BI, KPI Reporting* 2025
- Interactive Power BI dashboard over healthcare operational metrics, built for executive level KPI tracking and drill down analysis.

ACADEMIC PROJECTS

- Bridge Articles in the Wikipedia Knowledge Graph** | *EECS 4414 Information Networks, group of four* 2026
- Network analysis identifying bridge articles that connect otherwise separated topic communities in the Wikipedia link graph, covering centrality measures, community detection and result interpretation.
- Auto Insurance Data Warehouse and Hotel Booking Service** | *EECS 3421 Database Systems* 2026
- Designed a dimensional data warehouse for auto insurance analytics, then built a relational hotel booking service with a team of five covering schema design, normalization, constraints and query workloads.
- Microsoft and Netflix Financial Analysis** | *ENTR 3600, group of four* 2026
- Financial statement analysis and comparative valuation of two public technology companies, presented to the class.

LEADERSHIP, VOLUNTEERING AND COMMUNITY

Lead Developer and Executive

Oct. 2024 – Oct. 2025

York University Data Science Club

Toronto, ON

- Set technical direction for a 12 person team across 5+ machine learning projects, covering project scoping, code review and technical mentorship for members newer to Python and scikit-learn.
- Published 8 technical articles in Level Up Coding and CodeX while in the role.

Student Ambassador and Week Zero Leader

Summer 2024

Lassonde School of Engineering, York University

Toronto, ON

- Represented Lassonde to prospective and incoming students, and led Week Zero orientation programming for first year engineering and computer science cohorts.

Core Volunteer and Speaker

2023

Vadodara Startup Festival and University Tech Expo

Gujarat, India

- Core volunteer for the Vadodara Startup Festival (December 2023); organizer and speaker at a university Tech Expo.

PUBLICATIONS AND WRITING

The Awakening Code — published book, sole author.

Technical writing — author across 8 publications, including Level Up Coding and CodeX, covering applied machine learning, generative AI systems and software engineering practice.

CERTIFICATIONS

Microsoft Azure AI Essentials: Workloads and Machine Learning on Azure

Career Essentials in Generative AI, Microsoft and LinkedIn

Machine Learning Foundations: Linear Algebra

Introduction to Artificial Intelligence

Ethics in the Age of Generative AI

HONOURS AND AWARDS

Co-op of the Year Nominee, Ontario Government co-op program (2025)

5x University Gold Medalist, Parul University, awarded for highest GPA overall

Dean's List, Parul University

Published author and technical writer across 8 publications

LANGUAGES

English (full professional proficiency), Gujarati, Hindi. Currently studying French toward TEF Canada certification.